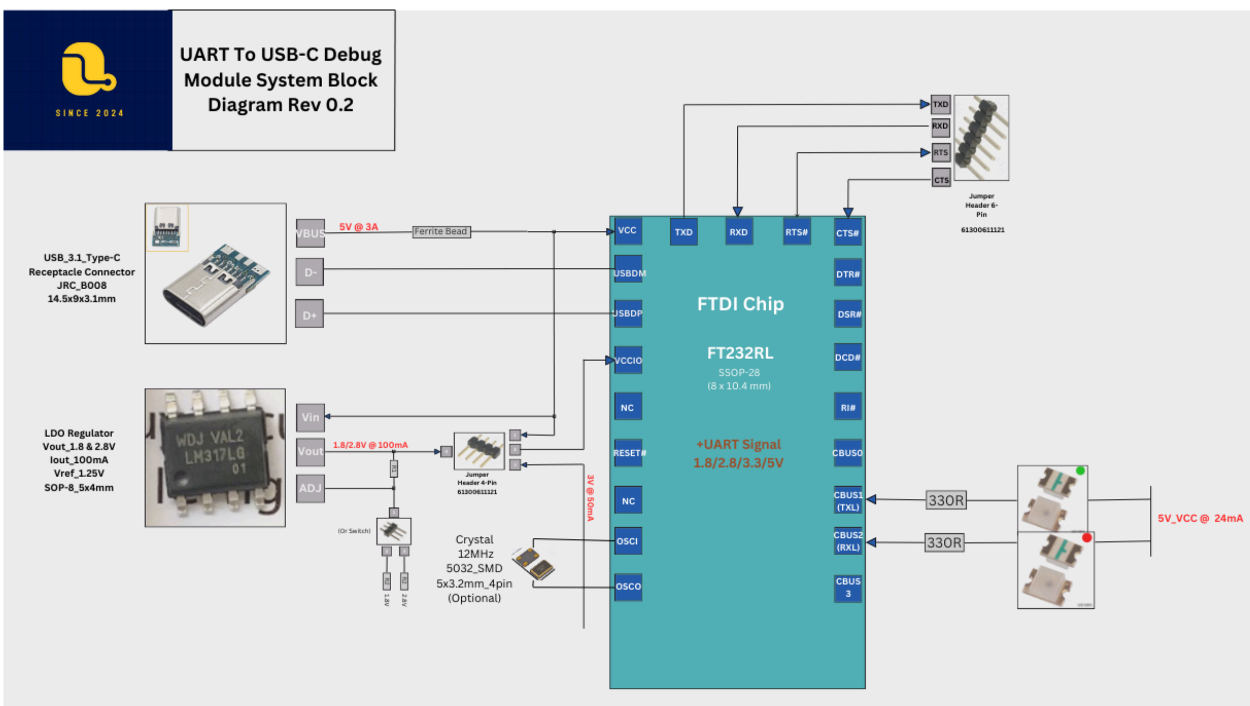
## Internship Project Report - Vita Imaging Company
## August 1, 2024

## Quang Nguyen

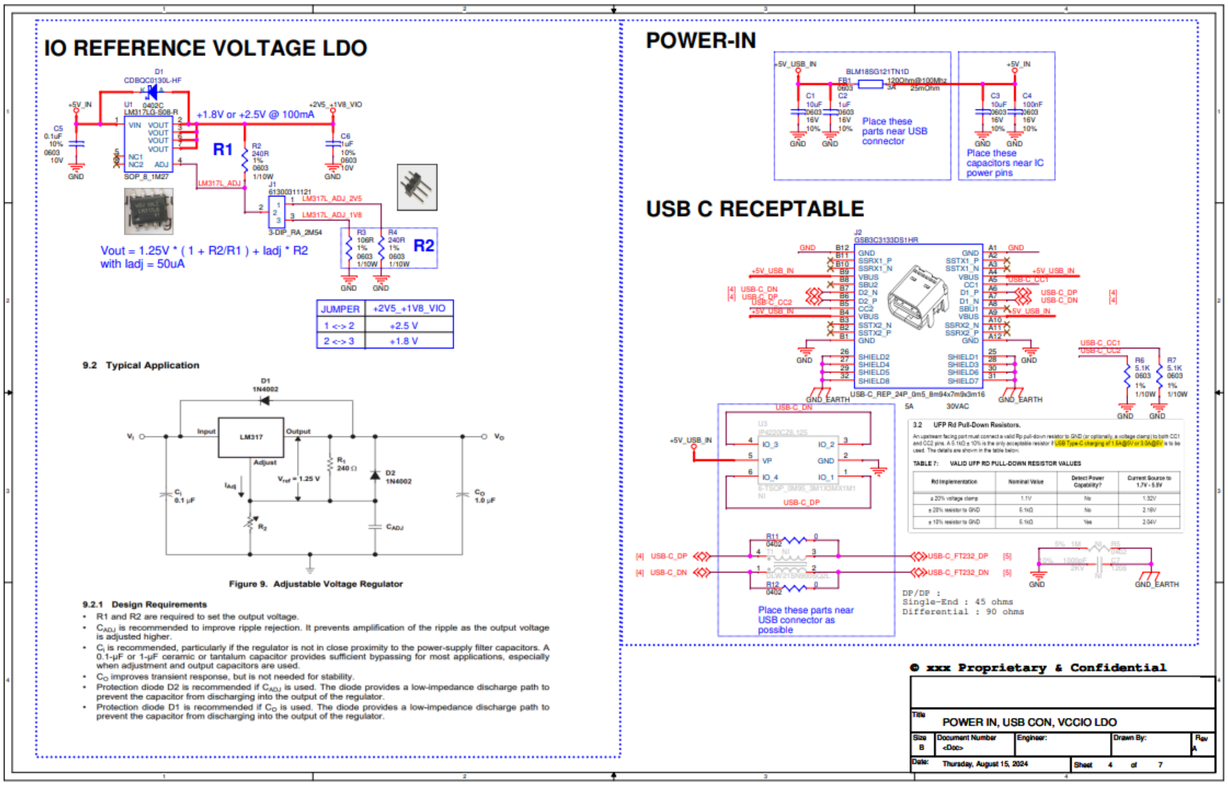
This PCB design is for a new USB-UART communication board, based on the FT232RL documentation. This circuit board provides a reliable method for data transfer between a host (e.g. computer) and various devices for programming and debugging. For accessibility, the USB 2.0 standard is implemented with a USB-C connector.

The circuit board primarily consists of an FT232RL USB UART IC chip (for the USB to serial UART interface, as well as providing connections for input power, input usb connections, UART output headers and a clock), and an LDO Regulator acting like a voltage divider to have the options of providing a lower voltage output of 1.8/2.2V (can be selected with a female-female jumper) from the input 5V for smaller power consumption. And, there are obviously many other components with varying types, such as capacitors, resistors, LEDs (indicating sending signals), crystal for clock synchronization. These are placed in accordance with the datasheets of the main chip and regulator.

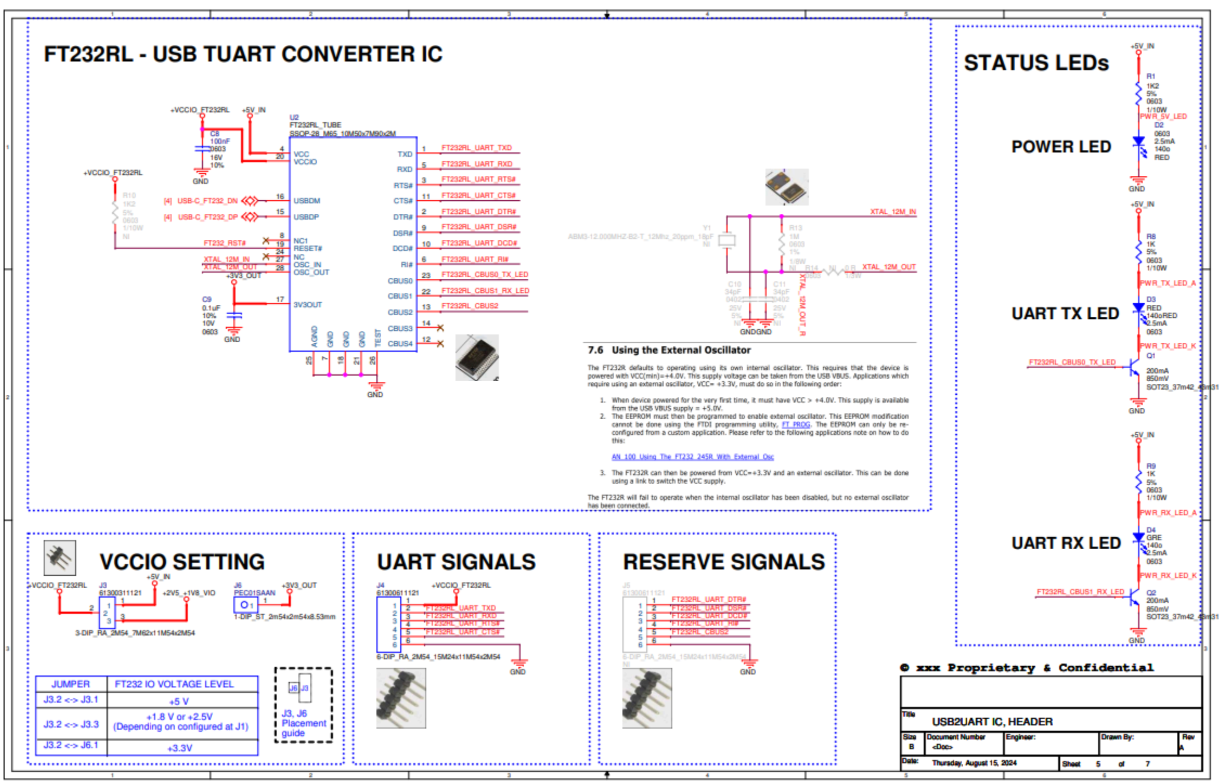



*System Block Diagram for the USC-UART Module*

I used **Orcad Capture** for the schematic drawings and **Allegro PCB designer** for the PCB board layout. These are provided with licenses, hence only a very limited access time is given to an intern and a student.



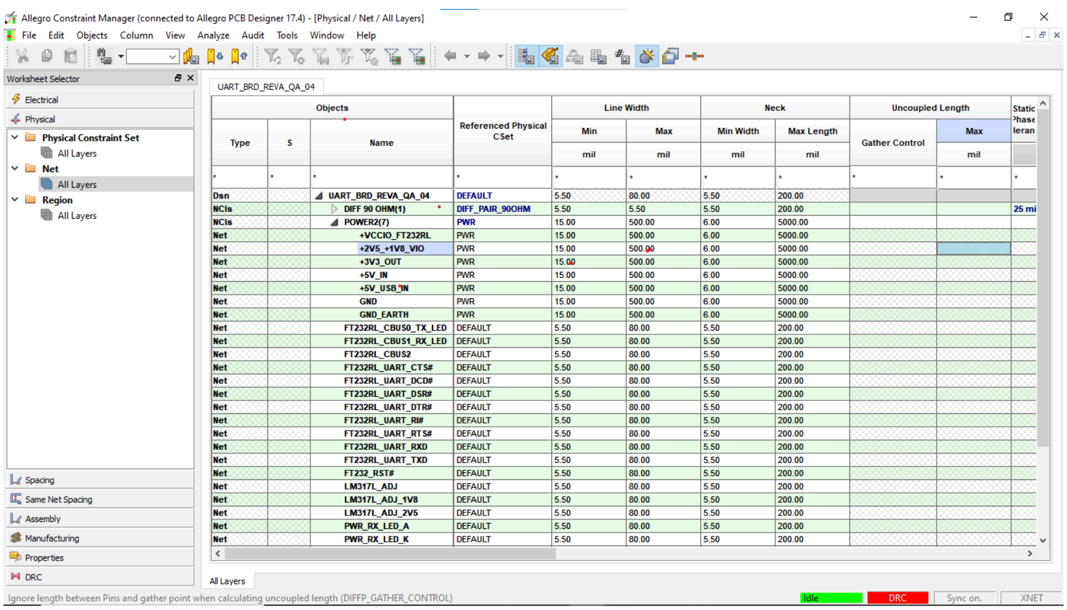

*Schematics for the USB connections, LDO connections, and the Power IN for the IC chip*




*Schematics for the IC chip connections, with LEDs, Headers for UART*

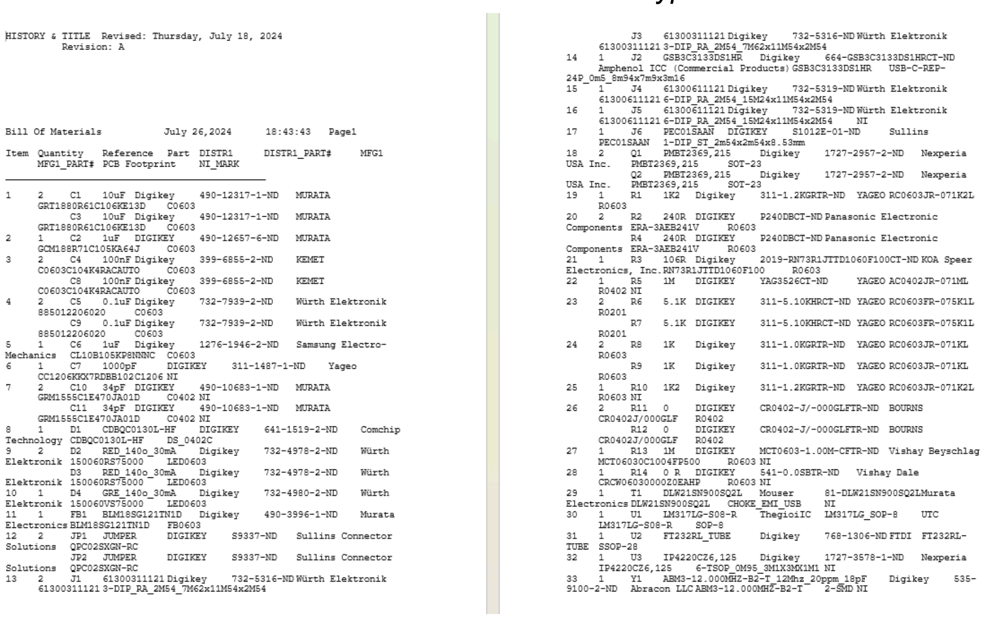
I also designed the footprints in the PADStack editor for each component as SMD pads with the dimensions and drills according to the design rules and the device datasheets. Then, importing the netlist into the PCB editor for placing, I simply made the connections with traces. Under the Constraints setting, the traces for power lines such as +5V_IN, +VCCIO_FT232RL, etc, are set for 15 mils for better heat dissipation as a larger flow current. The differential pairs for the USB-C DN/DP are also set for a slightly larger width of 6.5 mil with fixed spacing in between.

Since this is a small board with relatively few components, a 2-layer board is sufficient enough with the top layer containing all the connections for the components and the chips and the power plane, while the bottom layer contains the ground plane. There is no need for a separate layer for the power distribution plane.



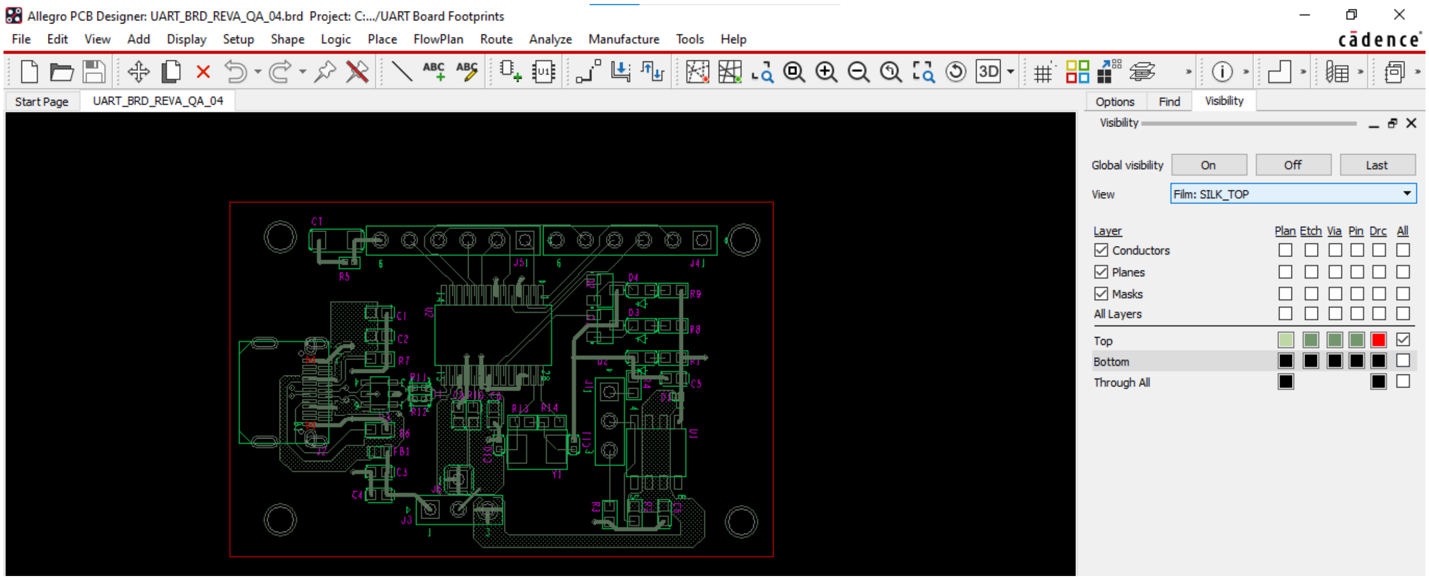

| Objects | | | Referenced Physical CSet | Line Width | | Neck | | Uncoupled Length | | Static Phase Toleran |
|---|---|---|---|---|---|---|---|---|---|---|
| Type | S | Name | | Min | Max | Min Width | Max Length | Gather Control | Max | |
| | | | | mil | mil | mil | mil | | mil | |
| * | * | * | * | * | * | * | * | * | * | * |
| Dsn | | UART_BRD_REVA_QA_04 | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| NCls | | DIFF 90 OHM(1) | DIFF_PAIR_90OHM | 5.50 | 5.50 | 5.50 | 200.00 | | | 25 mi |
| NCls | | POWER2(7) | PWR | 15.00 | 500.00 | 6.00 | 5000.00 | | | |
| Net | | +VCCIO_FT232RL | PWR | 15.00 | 500.00 | 6.00 | 5000.00 | | | |
| Net | | +2V5_+1V8_VIO | PWR | 15.00 | 500.00 | 6.00 | 5000.00 | | | |
| Net | | +3V3_OUT | PWR | 15.00 | 500.00 | 6.00 | 5000.00 | | | |
| Net | | +5V_IN | PWR | 15.00 | 500.00 | 6.00 | 5000.00 | | | |
| Net | | +5V_USB_IN | PWR | 15.00 | 500.00 | 6.00 | 5000.00 | | | |
| Net | | GND | PWR | 15.00 | 500.00 | 6.00 | 5000.00 | | | |
| Net | | GND_EARTH | PWR | 15.00 | 500.00 | 6.00 | 5000.00 | | | |
| Net | | FT232RL_CBUS0_TX_LED | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | FT232RL_CBUS1_RX_LED | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | FT232RL_CBUS2 | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | FT232RL_UART_CTS# | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | FT232RL_UART_DCD# | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | FT232RL_UART_DSR# | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | FT232RL_UART_DTR# | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | FT232RL_UART_RI# | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | FT232RL_UART_RTS# | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | FT232RL_UART_RXD | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | FT232RL_UART_TXD | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | FT232_RST# | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | LM317L_ADJ | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | LM317L_ADJ_1V8 | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | LM317L_ADJ_2V5 | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | PWR_RX_LED_A | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |
| Net | | PWR_RX_LED_K | DEFAULT | 5.50 | 80.00 | 5.50 | 200.00 | | | |

*Trace width constraints for different types of lines*

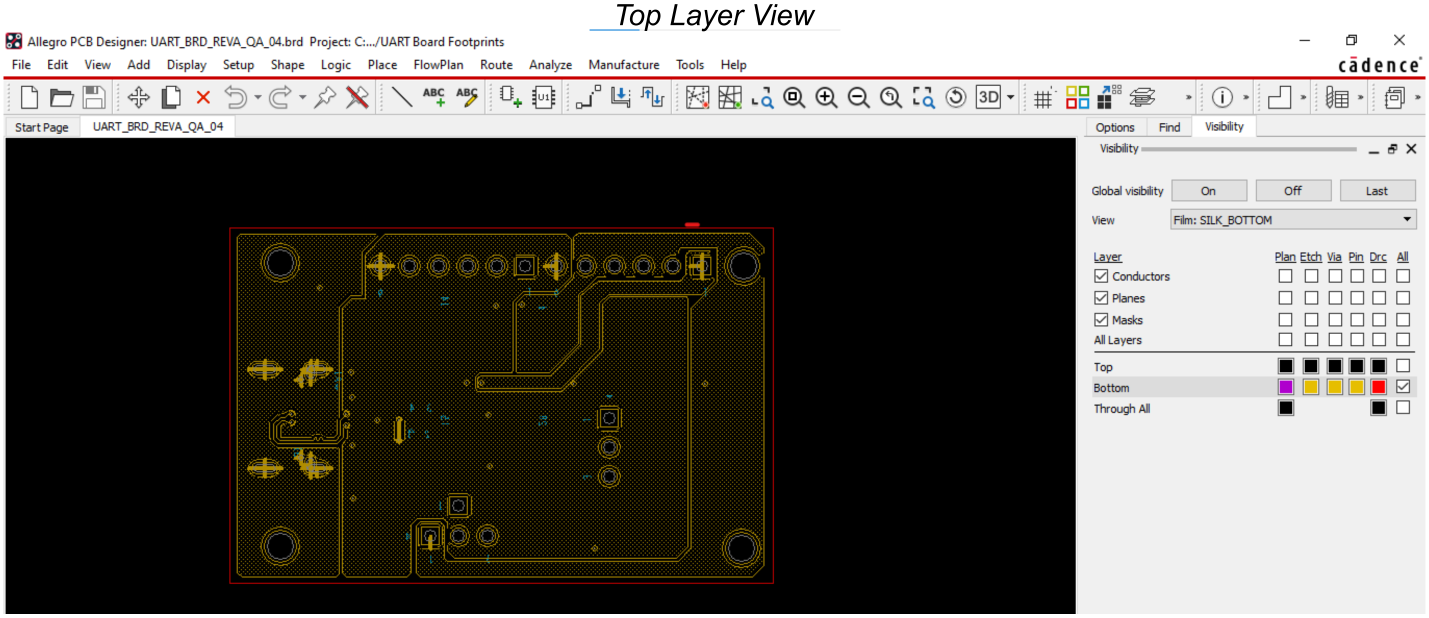


```
HISTORY & TITLE  Revised: Thursday, July 18, 2024
        Revision: A



Bill Of Materials          July 26,2024      18:43:43   Page1

Item  Quantity   Reference  Part  DISTR1      DISTR1_PART#     MFG1
      MFG1_PART# PCB Footprint    NI_MARK
_______________________________________________

1     2     C1    10uF  Digikey     490-12317-1-ND   MURATA
      GRT1880R61C106KE13D    C0603
            C3    10uF  Digikey     490-12317-1-ND   MURATA
      GRT1880R61C106KE13D    C0603
2     1     C2    1uF   DIGIKEY     490-12657-6-ND   MURATA
      GCM188R71C105KA64J     C0603
3     2     C4    100nF Digikey     399-6855-2-ND    KEMET
      C0603C104K4RACAUTO     C0603
            C8    100nF Digikey     399-6855-2-ND    KEMET
      C0603C104K4RACAUTO     C0603
4     2     C5    0.1uF Digikey     732-7939-2-ND    Würth Elektronik
      885012206020      C0603
            C9    0.1uF Digikey     732-7939-2-ND    Würth Elektronik
      885012206020      C0603
5     1     C6    1uF   Digikey     1276-1946-2-ND   Samsung Electro-
Mechanics   CL10B105KP8NNNC  C0603
6     1     C7    1000pF      DIGIKEY     311-1487-1-ND    Yageo
      CC1206KKX7RDBB102C1206 NI
7     2     C10   34pF  DIGIKEY     490-10683-1-ND   MURATA
      GRM1555C1E470JA01D     C0402 NI
            C11   34pF  DIGIKEY     490-10683-1-ND   MURATA
      GRM1555C1E470JA01D     C0402 NI
8     1     D1    CDBQC0130L-HF     DIGIKEY     641-1519-2-ND    Comchip
Technology CDBQC0130L-HF    DS_0402C
9     2     D2    RED_140o_30mA     Digikey     732-4978-2-ND    Würth
Elektronik 150060RS75000    LED0603
            D3    RED_140o_30mA     Digikey     732-4978-2-ND    Würth
Elektronik 150060RS75000    LED0603
10    1     D4    GRE_140o_30mA     Digikey     732-4980-2-ND    Würth
Elektronik 150060VS75000    LED0603
11    1     FB1   BLM18SG121TN1D    Digikey     490-3996-1-ND    Murata
Electronics BLM18SG121TN1D   FB0603
12    2     JP1   JUMPER      DIGIKEY     S9337-ND    Sullins Connector
Solutions   QPC02SXGN-RC
            JP2   JUMPER      DIGIKEY     S9337-ND    Sullins Connector
Solutions   QPC02SXGN-RC
13    2     J1    61300311121 Digikey     732-5316-ND Würth Elektronik
      61300311121 3-DIP_RA_2M54_7M62x11M54x2M54
            J3    61300311121 Digikey     732-5316-ND Würth Elektronik
      61300311121 3-DIP_RA_2M54_7M62x11M54x2M54
14    1     J2    GSB3C3133DS1HR    Digikey     664-GSB3C3133DS1HRCT-ND
      Amphenol ICC (Commercial Products) GSB3C3133DS1HR   USB-C-REP-
24P_0m5_8m94x7m9x3m16
15    1     J4    61300611121 Digikey     732-5319-ND Würth Elektronik
      61300611121 6-DIP_RA_2M54_15M24x11M54x2M54
16    1     J5    61300611121 Digikey     732-5319-ND Würth Elektronik
      61300611121 6-DIP_RA_2M54_15M24x11M54x2M54     NI
17    1     J6    PEC01SAAN   DIGIKEY     S1012E-01-ND     Sullins
      PEC01SAAN   1-DIP_ST_2m54x2m54x8.53mm
18    2     Q1    PMBT2369,215      Digikey     1727-2957-2-ND   Nexperia
USA Inc.    PMBT2369,215      SOT-23
            Q2    PMBT2369,215      Digikey     1727-2957-2-ND   Nexperia
USA Inc.    PMBT2369,215      SOT-23
19    1     R1    1K2   Digikey     311-1.2KGRTR-ND  YAGEO RC0603JR-071K2L
      R0603
20    2     R2    240R  DIGIKEY     P240DBCT-ND Panasonic Electronic
Components  ERA-3AEB241V      R0603
            R4    240R  DIGIKEY     P240DBCT-ND Panasonic Electronic
Components  ERA-3AEB241V      R0603
21    1     R3    106R  Digikey     2019-RN73R1JTTD1060F100CT-ND KOA Speer
Electronics, Inc.RN73R1JTTD1060F100     R0603
22    1     R5    1M    DIGIKEY     YAG3526CT-ND     YAGEO AC0402JR-071ML
      R0402 NI
23    2     R6    5.1K  DIGIKEY     311-5.10KHRCT-ND YAGEO RC0603FR-075K1L
      R0201
            R7    5.1K  DIGIKEY     311-5.10KHRCT-ND YAGEO RC0603FR-075K1L
      R0201
24    2     R8    1K    Digikey     311-1.0KGRTR-ND  YAGEO RC0603JR-071KL
      R0603
            R9    1K    Digikey     311-1.0KGRTR-ND  YAGEO RC0603JR-071KL
      R0603
25    1     R10   1K2   Digikey     311-1.2KGRTR-ND  YAGEO RC0603JR-071K2L
      R0603 NI
26    2     R11   0     DIGIKEY     CR0402-J/-000GLFTR-ND  BOURNS
      CR0402J/000GLF    R0402
            R12   0     DIGIKEY     CR0402-J/-000GLFTR-ND  BOURNS
      CR0402J/000GLF    R0402
27    1     R13   1M    DIGIKEY     MCT0603-1.00M-CFTR-ND  Vishay Beyschlag
      MCT06030C1004FP500     R0603 NI
28    1     R14   0 R   DIGIKEY     541-0.0SBTR-ND   Vishay Dale
      CRCW06030000Z0EAHP     R0603 NI
29    1     T1    DLW21SN900SQ2L    Mouser      81-DLW21SN900SQ2LMurata
Electronics DLW21SN900SQ2L   CHOKE_EMI_USB     NI
30    1     U1    LM317LG-S08-R     ThegioiIC   LM317LG_SOP-8    UTC
      LM317LG-S08-R     SOP-8
31    1     U2    FT232RL_TUBE      Digikey     768-1306-ND FTDI  FT232RL-
TUBE  SSOP-28
32    1     U3    IP4220CZ6,125     Digikey     1727-3578-1-ND   Nexperia
      IP4220CZ6,125     6-TSOP_0M95_3M1X3MX1M1 NI
33    1     Y1    ABM3-12.000MHZ-B2-T_12Mhz_20ppm_18pF    Digikey     535-
9100-2-ND   Abracon LLC ABM3-12.000MHZ-B2-T     2-SMD NI
```

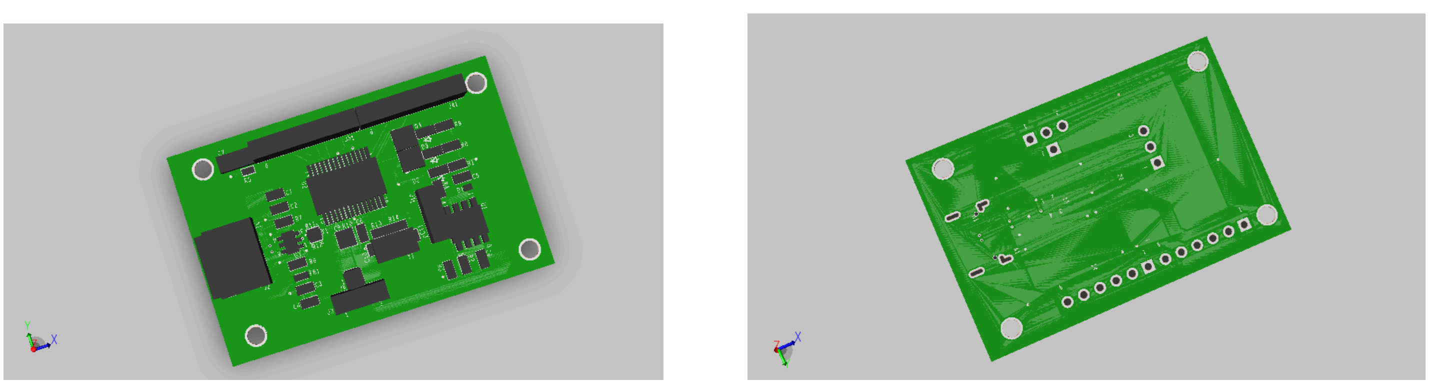
*BOM file for all components in the Schematics*




Top Layer View




Bottom Layer View



3D Top and Bottom Images of the board

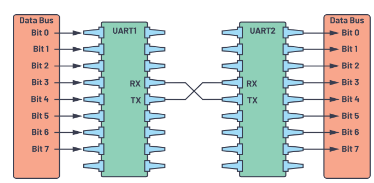
**Components and Design Considerations**

In designing this board, several hardware considerations were made to ensure reliability, signal integrity, and protection for both the USB interface and the FT232RL chip. One of the first things I added was ESD protection for the USB-C port. Since the USB connector is the most exposed part of the board, it is vulnerable to electrostatic discharge when users plug and unplug cables. To prevent any voltage spikes from reaching and damaging the FT232RL, I placed a set of low-capacitance TVS diodes close to the USB D+ and D– lines. These clamp high-voltage transients to safe levels while maintaining the high-speed nature of USB communication.

Right after the USB connector, I also included a ferrite bead in series with the 5V input line. Its job is to filter high-frequency noise coming from either the host or the board itself. Combined with a set of decoupling capacitors placed near the regulator and the FT232RL power pins, this helps stabilize the supply rails. The capacitors smooth out sudden current changes, while the ferrite bead blocks unwanted RF interference, so the board behaves predictably even during rapid data transfer.

For the USB differential pair (D+ and D–), I followed the routing rules recommended in the FT232RL and USB 2.0 documentation. These traces are length-matched and kept at a controlled spacing to maintain the characteristic impedance needed for USB full-speed operation. Keeping them as a differential pair helps reduce noise pickup and ensures a clean data signal between the connector and the FT232RL.

On the UART side, I made sure the TX and RX lines were routed cleanly and kept away from noisy digital edges. Since this board may be used for both programming and debugging, the UART header provides a straightforward 3- or 4-pin interface for connecting to a microcontroller or peripheral device. The FT232RL handles UART framing internally, but the board design ensures stable logic-level signals through proper decoupling, trace lengths, and reference grounding.




For power management, the voltage regulator plays a key role in allowing the board to operate with different output voltage options. Even though the FT232RL can work with 3.3V I/O, some devices require even lower voltages, so the regulator is configured to output selectable 1.8V or 2.2V using a jumper. This helps reduce overall power consumption when working with low-voltage digital circuits. The regulator layout follows the datasheet guidelines: input and output capacitors placed close to the pins, tight grounding, and a short path for the feedback loop to avoid oscillations.

*Personal Notes: I was greatly instructed by the hardware engineers at the company during this summer's internship. Hence, the design files follow the design rules and guidelines set by the engineers; however, there remain mistakes in the final file layout export, as I lack experience with making manufacturing quality files.*